American Journal of Transplantation 2004; 4: 905–913
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doi: 10.1111/j.1600-6143.2004.00450.x

Cancer after Kidney Transplantation in the United States

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Previous reports of cancer after kidney transplantation have been limited by small numbers of patients in single-center studies and incomplete ascertainment of cases in large registries.

We examined rates of malignancies among first-time recipients of deceased or living donor kidney transplantations in 1995–2001 (n = 35 765) using Medicare billing claims.

For most common tumors, e.g. colon, lung, prostate, stomach, esophagus, pancreas, ovary and breast, cancer rates were roughly twofold higher after kidney transplantation compared with the general population. Melanoma, leukemia, hepatobiliary tumors, cervical and vulvovaginal tumors were each approximately fivefold more common. Testicular and bladder cancers were increased approximately threefold, while kidney cancer was approximately 15-fold more common. Kaposi's sarcoma, non-Hogkin's lymphomas, and nonmelanoma skin cancers were more than 20-fold increased than in the general population. Compared with patients on the waiting list, several tumors were more common after transplantation ($p < 0.01$): nonmelanoma skin cancers (2.6-fold), melanoma (2.2-fold), Kaposi's sarcoma (9.0-fold), non-Hodgkin's lymphoma (3.3-fold), cancer of the mouth (2.2-fold), and cancer of the kidney (39% higher).

The rates for most malignancies are higher after kidney transplantation compared with the general population. Cancer should continue to be a major focus of prevention in kidney transplantation.

Key words: Hemodialysis, malignancies, United Network for Organ Sharing, United States Renal Data System, waiting list

Received 25 October 2003, revised and accepted for publication 13 February 2004

Introduction

Long-term complications of kidney transplantation have assumed increasing importance as short-term patient and graft survival have improved. There are reasons to believe that cancer should be more common after kidney transplantation than in the general population, and more common than in comparable patients on dialysis. Immunosuppressive agents may cause DNA damage and interfere with normal DNA repair mechanisms. In addition, a number of cancers have been linked to viral infections that are common after transplantation. Finally, immune surveillance, which ordinarily prevents the growth and development of malignancies, may be impaired by immunosuppressive medications.

It is difficult to precisely ascertain the incidence of most tumors, and to compare their rates of occurrence with those in the general population, using data from small, single-center studies. There have been few reports on the incidence of cancer from transplant registries (1–5). Reporting of cancer to registries is often incomplete, and the extent that registry data underestimate the true incidence of cancer is difficult to determine.

We linked data from the United States Renal Data System (USRDS) to Medicare billing claims to detect the occurrence of cancers after kidney transplantation. In the USRDS 2003 Annual Data Report, we reported the 3-year incidences of most major post-transplant malignancies, some of the clinical correlates to cancer after transplantation, and the associations between cancer and outcomes (6). In the present report, we compare sex-specific rates of cancer after transplantation with those in the general US population, adjusting for age. We also compare rates of cancer in kidney transplant recipients with rates among comparable patients on the deceased donor waiting list.

Methods

Ascertaining the occurrence of cancer

To identify patients with cancer, Medicare claims were searched for the appropriate ICD-9-CM codes (Appendix). Patients were classified as having a specific malignancy if one inpatient (Part A) or two outpatient/Part B Medicare claims were found within 1 year of each other. This method has been validated for identifying diabetes (7). To validate this technique for cancers, we compared Medicare claims with the results of a survey of Medicare beneficiaries conducted in 1992. Using one Part A or two outpatient/Part B

claims within 1 year predicted the results of the survey with reasonable positive predictive values (PPV; the proportion of claimants found to have cancer in the survey) and negative predictive values (NPV; the proportion without claims found not to have cancer in the survey). For skin cancer these were 81% (PPV) and 85% (NPV), for prostate in men 61% and 97%, for lung 71% and 100%, for breast in women 87% and 95%, for kidney 79% and 100%, for ovarian in women 40% and 99%, for stomach 67% and 99%, for uterine in women 50% and 97%, for cervical in women 43% and 99%, and for bladder 72% and 99%, respectively. To ensure a high likelihood that cancer would be detected, patients were censored at either: (1) loss of Medicare primary coverage, (2) 3 years after transplantation, (3) 12/31/2001, or (4) death.

Determining the incidence of cancer after transplantation

For this analysis, we used first-time recipients of deceased or living donor kidney transplantations between 1995 and 2001 with known age at the time of transplantation. Patients with other organ transplants were excluded. All patients had Medicare as the primary payer at the time of transplantation. This latter determination was made by first identifying that Medicare was the primary payer for the transplantation itself, through Medicare billing data. Then the Medicare enrollment database was searched to determine Medicare Part A and Part B primary pay status along with coverage start and stop dates. There were 35 765 (47%) fulfilling these criteria out of a total of 76 467 first transplantations between 1995 and 2001.

Three-year, cumulative incidences were estimated from a multivariate Cox proportional hazards analysis adjusting for variables recorded at the time of transplantation: age, gender, race, Hispanic ethnicity, primary cause of renal failure, year of transplantation, donor type, hepatitis B & C serology, education level, employment status, time on dialysis, donor age, donor gender, donor race, HLA mismatches, recipient–donor body surface area matching, body mass index, panel reactive antibodies, cytomegalovirus matching, baseline maintenance immunosuppression (cyclosporine, microemulsion cyclosporine, tacrolimus, sirolimus, azathioprine, mycophenolate mofetil), as well as monoclonal, polyclonal, and anti-interleukin-2 receptor antibodies for induction.

Comparing cancer after transplantation with cancer in the US population

Estimates of cancer incidence rates in the general US population were obtained from the United States Cancer Statistics: 1999 Incidence report published by the United States Cancer Statistics Working Group (8). Crude first-, second-, and third-year post-transplantation cancer incidence rates were calculated for male and female recipients, and rates from both the general population and the transplantation population were age-adjusted to the 2000 census population. For each tumor type we calculated a rate ratio by dividing the incidence rate for the transplant patients by the rate for the general population. Statistical tests of differences were not calculated.

Comparing cancer after transplantation with the waiting list

To calculate the relative risk for cancer associated with transplantation vs. comparable end-stage kidney disease patients without transplantation, we compared recipients with patients who were on the waiting list. This analysis included 42 201 patients who were placed on the United Network for Organ Sharing list for deceased donor kidney transplantation. Of these, 20 183 were listed and ultimately transplanted (with either a deceased donor or a living donor kidney), while 22 018 remained on the waiting list without receiving a transplant. In addition, we included 3905 patients who were preemptively transplanted, i.e. before being placed on the list. Altogether there were 46 106 patients included in this analysis. This analysis was also restricted to those patients who were Medicare primary pay at the time of listing or preemptive transplantation.

The relative risk of cancer after transplantation compared with the waiting list was assessed using a time-dependent Cox proportional hazards analysis. Patients were censored at the loss of Medicare primary payer status, death, 3 years post-transplant, or on December 31, 2001. Transplantation status was considered a time-dependent covariate. Patients who were preemptively transplanted were included in the transplantation group from the initiation of their follow-up period.

Results

Incidence and clinical correlates of cancer after kidney transplantation

The cumulative incidence of any nonskin malignancy (but including melanoma) after kidney transplantation was 1.2%, 1.9%, 3.3%, 5.5%, and 7.5% at months 3, 6, 12, 24, and 36, respectively (Table 1). The cumulative incidence of skin cancer (excluding melanoma) was 0.3%, 0.9%, 2.3%, 5.0%, and 7.4% at months 3, 6, 12, 24, and 36 months (Table 1).

Table 1: Cumulative incidences of malignancies at 1, 2 and 3 years after transplantation (%)

Type of malignancy	1 year	2 year	3 year
Skin			
Skin	2.25	4.95	7.43
Melanoma	0.09	0.19	0.32
Gastrointestinal			
Colon	0.18	0.33	0.51
Esophagus	0.02	0.03	0.07
Hepatobiliary	0.06	0.16	0.22
Pancreas	0.03	0.07	0.10
Small intestine	0.01	0.02	0.03
Stomach	0.05	0.08	0.11
Genitourinary			
Bladder	0.16	0.26	0.32
Cervix (women)	0.02	0.09	0.18
Kidney	0.54	0.82	1.01
Ovary (women)	0.04	0.09	0.14
Prostate (men)	0.70	1.28	1.74
Testes (men)	0.02	0.04	0.06
Uterus (women)	0.07	0.13	0.18
Vulvovaginal (women)	0.03	0.05	0.09
Lymphomas			
Hodgkin's	0.04	0.05	0.08
Non-Hodgkin's	0.53	0.81	1.02
Other			
Bone	0.04	0.06	0.10
Breast (women)	0.46	0.80	1.05
Breast (men)	0.01	0.04	0.05
CNS	0.09	0.20	0.29
Endocrine	0.09	0.17	0.21
Kaposi's sarcoma	0.08	0.12	0.14
Larynx	0.02	0.03	0.07
Leukemia	0.14	0.19	0.25
Lung	0.22	0.41	0.69
Mouth	0.13	0.22	0.32
Myeloma	0.10	0.15	0.19
Other	0.66	1.43	2.10
Any non-skin	3.27	5.52	7.45

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Table 2: Major risk factors for malignancies after kidney transplantation*

	Non-skin malignancy		Non-melanoma skin	
Risk factor	RR (95% CI)	p-value	RR (95% CI)	p-value
Year of transplantation				
1999–01 vs. 1995–98	1.04 (0.92–1.18)	0.5048	1.04 (0.91–1.19)	0.5976
Age at transplantation				
0–17 years	1.62 (1.17–2.24)	0.0036	0.15 (0.02–1.09)	0.0604
18–34 years (reference)	1.00	–	1.00	–
35–49 years	1.41 (1.18–1.68)	0.0002	4.31 (2.99–6.22)	<0.0001
50–64 years	2.98 (2.51–3.52)	<0.0001	15.0 (10.3–20.8)	<0.0001
65 years and older	4.93 (4.10–5.94)	<0.0001	27.1 (19.0–38.8)	<0.0001
Sex				
Men vs. women	1.20 (1.09–1.32)	0.0002	2.18 (1.94–2.45)	<0.0001
Race				
Caucasian (reference)	1.00	–	1.00	–
African American	0.84 (0.75–0.95)	0.0062	0.06 (0.04–0.08)	<0.0001
Asian	0.65 (0.49–0.85)	0.0021	0.11 (0.06–0.20)	<0.0001
Other	0.67 (0.47–0.95)	0.0243	0.29 (0.16–0.51)	<0.0001
Ethnicity				
Hispanic vs. non-Hispanic	0.75 (0.64–0.87)	0.0002	0.29 (0.23–0.36)	<0.0001
Primary cause of renal failure				
Diabetes	0.82 (0.71–0.95)	0.0077	0.63 (0.54–0.73)	<0.0001
Glomerulonephritis (reference)	1.00	–	1.00	–
Hypertension	1.06 (0.92–1.22)	0.4190	1.05 (0.91–1.21)	0.5253
Cystic kidney disease	0.99 (0.82–1.19)	0.8798	1.27 (1.08–1.48)	0.0033
Other	1.49 (1.31–1.69)	<0.0001	0.89 (0.77–1.03)	0.1107
Dialysis before transplantation				
None (pre-emptive)	0.83 (0.62–1.10)	0.1896	1.10 (0.88–1.38)	0.4064
Up to 1 year (reference)	1.00	–	1.00	–
1 up to 2 years	0.97 (0.82–1.14)	0.7234	0.86 (0.74–1.00)	0.0477
2 up to 3 years	1.12 (0.95–1.31)	0.1961	0.84 (0.72–0.99)	0.0334
3 or more years	1.20 (1.03–1.40)	0.0222	0.79 (0.68–0.92)	0.0030
Body mass index				
<18.5 kg/m^2	1.03 (0.81–1.31)	0.8336	1.16 (0.86–1.57)	0.3227
18.5–24.9 kg/m^2 (reference)	1.00	–	1.00	–
25.0–29.9 kg/m^2	0.97 (0.87–1.08)	0.5602	0.82 (0.73–0.92)	0.0005
30.0–34.9 kg/m^2	0.89 (0.78–1.03)	0.1227	0.72 (0.61–0.83)	<0.0001
≥ 35.0 kg/m^2	0.98 (0.79–1.20)	0.8122	0.59 (0.45–0.78)	0.0002
Initial maintenance immunosuppression				
Cyclosporine	1.01 (0.86–1.18)	0.9268	1.02 (0.86–1.21)	0.8146
Microemulsion cyclosporine	0.98 (0.85–1.13)	0.7428	1.01 (0.86–1.18)	0.9110
Tacrolimus	0.94 (0.80–1.10)	0.4435	0.65 (0.54–0.79)	<0.0001
Sirolimus	0.89 (0.70–1.13)	0.3260	0.76 (0.56–1.02)	0.0659
Azathioprine	1.03 (0.90–1.20)	0.6494	1.17 (1.01–1.37)	0.0408
Mycophenolate mofetil	0.96 (0.84–1.09)	0.5111	1.03 (0.89–1.19)	0.6627
Antibody induction				
None (reference)	1.00	–	1.00	–
IL–2 receptor antibody	0.96 (0.83–1.12)	0.6162	0.85 (0.71–1.00)	0.0521
Other antibody	1.02 (0.92–1.14)	0.7165	0.83 (0.73–0.93)	0.0021

*Cox proportional hazards analysis, adjusted for living (v. deceased donor), hepatitis B and C serologies, education, employment, donor age, donor race, major histocompatibility mismatches, and panel reactive antibodies.
RR = relative risk, CI = confidence interval, IL-2 = interleukin-2.

In a Cox proportional hazards analysis, independent risk factors for nonskin malignancy after transplantation included older age, male gender, White and non-Hispanic ethnicity, and diabetes as the cause of kidney disease (Table 2). Interestingly, there was no difference in the incidence of malignancies for transplantations carried out from 1999–2001 compared with 1995–1998 (Table 2). The type of immunosuppression at the time of initial discharge after transplantation (analyzed by intention-to-treat) did not affect the risk of nonskin malignancy.

In general, risk factors for nonmelanoma skin cancers were similar to those for nonskin cancer (Table 2). However, there were some differences: obesity was associated with less risk for skin cancer, and a longer time on

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Table 3: Comparison of age-adjusted cancer rates in transplantation vs. the US population

	Cancer rates in men				Cancer rates in women			
		Year post-transplant				Year post-transplant		
Site	In the US pop.	Year 1	Year 2	Year 3	In the US pop.	Year 1	Year 2	Year 3
Skin								
Skin	24.0	2017.1	2333.3	2160.2	14.3	851.6	1306.8	1320.5
Melanoma	19.0	60.4	77.5	131.3	12.1	99.9	58.4	63.5
Gastrointestinal								
Colon	66.4	137.2	99.8	107.7	48.5	91.1	160.8	137.0
Esophagus	8.8	17.4	12.4	21.3	2.2	3.6	0.0	6.2
Hepatobiliary	9.4	33.5	143.1	39.2	5.4	83.2	11.8	24.3
Pancreas	12.3	19.8	34.1	12.4	9.6	25.1	19.7	44.1
Small intestine	2.0	3.8	7.8	4.1	1.4	25.1	0.0	0.0
Stomach	11.0	38.9	23.2	4.1	5.1	23.7	4.7	21.4
Genitourinary								
Bladder	38.3	148.9	80.2	60.9	10.0	69.0	36.3	36.3
Cervix	–	–	–	–	9.4	9.4	56.7	53.7
Kidney	16.0	671.0	236.3	226.1	8.4	767.7	136.1	122.9
Ovary	–	–	–	–	16.2	29.5	25.1	42.4
Prostate	162.0	477.4	360.5	265.8	–	–	–	–
Testes	5.5	21.3	16.3	20.4	–	–	–	–
Uterus	–	–	–	–	0.7	51.7	32.2	21.5
Vulvovaginal	–	–	–	–	3.0	14.6	14.2	26.5
Lymphomas								
Hodgkin's	3.2	37.9	12.4	98.6	2.5	11.5	0.0	93.5
Non-Hodgkin's	22	882.0	345.1	150.7	15.7	667.5	337.5	456.7
Other								
Bone	1.1	68.3	15.5	94.0	0.8	23.5	11.8	15.2
Breast	1.5	6.8	16.3	6.0	134.1	343.4	262.5	144.3
CNS	7.9	85.9	119.1	62.6	5.7	153.7	130.9	156.9
Endocrine	4.3	76.6	48.2	10.2	10.4	100.3	170.0	153.3
Larynx	7.9	15.1	6.2	36.3	1.7	8.9	24.5	21.0
Leukemia	14.5	102.0	41.1	28.1	8.8	157.0	39.3	69.7
Lung	89.1	149.4	155.5	202.8	53.4	141.8	87.7	194.1
Mouth	15.8	269.4	62.1	71.2	6.3	138.4	59.1	70.2
Kaposi's sarcoma	1.5	55.0	45.0	26.1	0.1	56.0	11.8	6.2

*Rates per 100 000 population for US estimates, and per 100 000 person-years for transplant recipients. All rates standardized to the 2000 US census population.

dialysis before transplantation was also associated with a lower risk for skin cancer. Among the types of initial immunosuppressive agents used, tacrolimus and antibody induction were associated with less risk for developing skin cancer, while azathioprine was associated with a higher risk for skin cancer (Table 2).

Comparing cancer after transplantation with cancer in the US population

For most common tumors, e.g. colon, lung, prostate, stomach, esophagus, pancreas, ovary and breast, cancer rates were roughly twofold higher after kidney transplantation compared with the general population (Tables 3 and 4). Melanoma, leukemia, hepatobiliary tumors, cervical and vulvovaginal tumors were each approximately fivefold more common. Testicular and bladder cancers were increased approximately threefold, while kidney cancer was approximately 15-fold more common than in the general population. Kaposi's sarcoma, non-Hodgkin's lymphomas, and nonmelanoma skin cancers were more than 20-fold increased compared with the general population (Tables 3 and 4).

For most cancers, the relative rates of increase over the general population were similar in men and women, except that tumors of the central nervous system and Kaposi's sarcoma were higher in women than men (Tables 3 and 4). In general, the relative rates of increase over the general population were also similar comparing post-transplantation years 1, 2, and 3. Possible exceptions included non-Hodgkin's lymphoma, leukemia, Kaposi's sarcoma, cancer of the mouth and kidney cancers, with each being higher in the first year compared with years 2 and 3 (Tables 3 and 4).

Comparing cancer after transplantation with the waiting list

Compared with patients on the waiting list, recipients had similar rates for most malignancies (Table 5). However, nonmelanoma skin cancers (2.6-fold higher), melanoma

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Table 4: Age-adjusted rate ratios for cancer in transplantation recipients compared with the US population, by year after transplantation

	Rate ratio* in men			Rate ratio* in women		
Site	Year 1	Year 2	Year 3	Year 1	Year 2	Year 3
Skin						
Non-melanoma	84.0	97.2	90.0	59.6	91.4	92.3
Melanoma	3.2	4.1	6.9	8.3	4.8	5.2
Gastrointestinal						
Colon	2.1	1.5	1.6	1.9	3.3	2.8
Esophagus	2.0	1.4	2.4	1.6	0.0	2.8
Hepatobiliary	3.6	15.2	4.2	15.4	2.2	4.5
Pancreas	1.6	2.8	1.0	2.6	2.1	4.6
Small intestine	1.9	3.9	2.1	17.9	0.0	0.0
Stomach	3.5	2.1	0.4	4.6	0.9	4.2
Genitourinary						
Bladder	3.9	2.1	1.6	6.9	3.6	3.6
Cervix	–	–	–	1.0	6.0	5.7
Kidney	41.9	14.8	14.1	91.4	16.2	14.6
Ovary	–	–	–	1.8	1.5	2.6
Prostate	2.9	2.2	1.6	–	–	–
Testes	3.9	3.0	3.7	–	–	–
Uterus	–	–	–	73.9	46.0	30.7
Vulvovaginal	–	–	–	4.9	4.7	8.8
Lymphomas						
Hodgkin's	11.8	3.9	30.8	4.6	0.0	37.4
Non-Hodgkin's	40.1	15.7	6.9	42.5	21.5	29.1
Other						
Bone	62.1	14.1	85.5	29.4	14.8	19.0
Breast	4.5	10.9	4.0	2.6	2.0	1.1
CNS	10.9	15.1	7.9	27.0	23.0	27.5
Endocrine	17.8	11.2	2.4	9.6	16.3	14.7
Larynx	1.9	0.8	4.6	5.2	14.4	12.4
Leukemia	7.0	2.8	1.9	17.8	4.5	7.9
Lung	1.7	1.7	2.3	2.7	1.6	3.6
Mouth	17.1	3.9	4.5	22.0	9.4	11.1
Kaposi's sarcoma	36.7	30.0	17.4	560.0	118.0	62.0

*Rate ratio is the age-adjusted rate in transplant recipients divided by the rate in the general population.

Table 5: Relative risk of cancer after transplantation compared with cancer while on the waiting list*

Type of cancer	RR (95% CI)	p-value
Skin	2.55 (2.26-2.88)	<0.0001
Melanoma	2.19 (1.31-3.65)	0.0028
Any non-skin	1.17 (1.07-1.28)	0.0004
Mouth	2.19 (1.33-3.61)	0.0022
Larynx	0.72 (0.28-1.83)	0.4886
Any gastrointestinal	0.83 (0.65-1.05)	0.1126
Esophagus	2.76 (1.03-7.37)	0.0428
Stomach	0.89 (0.45-1.78)	0.7436
Small intestine	0.43 (0.13-1.40)	0.1613
Colon	0.75 (0.54-1.04)	0.0860
Hepatobiliary	0.77 (0.46-1.31)	0.3430
Pancreas	0.68 (0.37-1.25)	0.2179
Lung	1.05 (0.79-1.40)	0.7241
Bone	0.91 (0.40-2.04)	0.8145
Breast in women	0.82 (0.57-1.17)	0.2648
Breast in men	1.88 (0.32-10.91)	0.4834
Kaposi's sarcoma	9.03 (2.58-31.60)	0.0005
Any genitourinary in women**	1.16 (0.86-1.56)	0.3425
Uterus	0.96 (0.43-2.16)	0.9259
Cervix	1.28 (0.48-3.36)	0.6230
Ovary	0.34 (0.12-0.97)	0.0439
Vulvovaginal	2.19 (0.67-7.12)	0.1936
Any genitourinary in men***	1.02 (0.86-1.21)	0.8592
Prostate	0.79 (0.62-1.00)	0.0460
Testes	0.61 (0.16-2.25)	0.4558
Bladder	1.12 (0.73-1.70)	0.6098
Kidney	1.39 (1.10-1.76)	0.0058
Central nervous system	1.27 (0.78-2.06)	0.3304
Endocrinologic	0.71 (0.45-1.12)	0.1433
Any hematopoietic	2.04 (1.64-2.53)	<0.0001
Non-Hogkin's lymphoma	3.29 (2.40-4.51)	<0.0001
Hodgkin's	2.60 (1.01-6.68)	0.0471
Myeloma	0.92 (0.57-1.49)	0.7338
Leukemia	1.59 (1.03-2.45)	0.0355
Other malignancies	1.06 (0.89-1.25)	0.5160

*Cox proportional hazards analysis with transplantation as a time-dependent covariate; adjusted for age, gender, race/ethnicity, primary cause of kidney failure, and prior duration of end-stage kidney disease.
**Includes uterus, cervix, ovary, vulvovaginal, bladder, and kidney.
***Includes prostate, testes, bladder, and kidney.
RR = relative risk, CI = confidence interval.

(2.2-fold higher), Kaposi's sarcoma (9.0-fold higher), non-Hodgkin's lymphoma (3.3-fold higher), cancer of the mouth (2.2-fold higher), and cancer of the kidney (39% higher) were all more common after transplantation ($p < 0.01$) compared with the waiting list (Table 5). Leukemia, Hodgkin's disease and esophageal cancers were also increased after transplantation ($0.01 < p < 0.05$) compared with the waiting list (Table 5). Interestingly, ovarian cancer and prostate cancer were less frequent after transplantation ($0.01 < p < 0.05$) compared with the waiting list.

Discussion

Studies in the general population have led to the development of guidelines for cancer screening and prevention (9–12). Whether or not these guidelines are applicable to kidney transplantation recipients depends on the life-expectancy of the individual patient and the risk of developing cancer (13). The present results suggest that the incidences of the most common cancers (colon, lung, prostate and breast) are roughly twofold higher in the first 3 years after kidney transplantation than in the general population. This suggests that measures to reduce the risk of these malignancies, e.g. screening for colon cancer, smoking cessation, screening for prostate and breast cancer, might also be appropriate for transplant recipients whose life expectancy is not several-fold less than individuals from the general population.

Single-center studies are generally too small to compare the rates of cancer after transplantation with those in the general population, and few registry studies have

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Table 6: Incidence of cancer in transplantation registries vs. the general population

	EDTA–ERA*	CTS	ANZ	DCR	SCR	USRDS
Number studied	NA	25 914	8618	1821	5931	35,765
Date of transplant	NA	1983–93	1965–97	NA	1970–97	1995–01
Transplanted organs included	K	K, H	K	K	K (84%) LI, H, L, P	K
Other population characteristics	Diagnosis in 1985–89	Women only	Survived ≥3 months			Medicare patients
Skin						
Non-melanoma	↑↑↑	↑↑↑	↑↑↑	↑↑↑	↑↑↑	↑↑↑
Melanoma	–	↑	NA	–	↑	↑↑
Gastrointestinal						
Colon	↑	NA	↑	NA	↑	↑
Esophagus	NA	NA	↑↑	NA	↑	↑
Hepatobiliary	↑	NA	↑↑	NA	–	↑↑
Pancreas	NA	NA	↑	NA	–	↑
Small Intestine	NA	NA	↑	NA	–	↑
Stomach	–	NA	↑	NA	↑	↑
Genitourinary						
Bladder	–	NA	↑↑	↑	↑	↑
Cervix	↑	NA	↑	NA	↑	↑
Kidney	NA	NA	↑↑	↑↑	↑↑	↑↑↑
Ovary	–	NA	–	NA	–	↑
Prostate	NA	NA	–	NA	–	↑
Testes	NA	NA	–	NA	–	↑
Uterus	–	NA	–	NA	–	↑↑↑
Vulvovaginal	NA	NA	↑↑↑	NA	↑↑↑	↑↑
Lymphomas						
Hodgkin's	NA	NA	↑	↑	–	↑↑
Non-Hodgkin's	↑↑	↑↑	↑↑	↑↑	↑↑	↑↑↑
Other						
Bone	NA	NA	–	NA	NA	↑↑↑
Breast	–	†	–	↑	–	↑
CNS	NA	NA	–	–	–	↑↑
Endocrine	NA	NA	↑	–	↑	↑↑
Larynx	NA	NA	↑	NA	–	↑
Leukemia	↑	NA	↑	NA	–	↑
Lung	–	↑	↑	↑	↑	↑
Mouth	NA	↑	↑	↑↑	↑↑	↑↑

*Rate differences were not quantified, and are therefore qualitative only.
Rates compared with the general population: †decreased, – not different, ↑ increased 2- to 4-fold, ↑↑ increased 5- to 19-fold, ↑↑↑ increased 20-fold.
NA = not available, EDTA–ERA = European Dialysis and Transplantation Association–European Renal Association (1), CTS = Collaborative Transplant Study (2), ANZ = Australia/New Zealand registry (3), DCR = Danish Cancer Registry (4), SCR = Swedish Cancer Registry (5), USRDS = United States Renal Data System; K = kidney, LI = liver, H = heart, L = lung, P = pancreas.

compared the incidence of cancer in kidney transplant recipients with the general population (1–5). The results of these registry studies are in general agreement with those of the current study (Table 6). However, in the current study the increase in the incidence of uterine cancer, compared with the general population, was much higher than that in other studies, and it is possible that some of these were coded as cervical cancer in other studies. Why the increased incidence of bone cancer is higher in the current study compared with other studies is not clear, although in some cases metastatic disease could have been miscoded as bone cancer.

We were unable to find any correlations (by intention-to-treat analysis) between the types of initial

Table 7: Initial immunosuppression

Agents	n	%
Maintenance		
Cyclosporine	5995	17
Microemulsion cyclosporine	16 985	47
Tacrolimus	9517	27
Sirolimus	2241	6
Azathioprine	7231	20
Mycophenolate mofetil	22 052	62
Antibody induction		
IL–2 receptor antibody	6448	18
Other antibody	8424	24

IL-2 = interleukin-2.

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immunosuppressive agents used and nonskin malignancies (Table 2). However, we did not examine the effects of these agents on individual tumors, and others have reported that antibody induction was associated with post-transplant lymphomas (14,15). Although there are theoretical reasons for believing that sirolimus might be associated with a reduced incidence of cancer (16), neither sirolimus, nor any other agent, were associated with nonskin cancers. However, only 6% of the study population were treated with sirolimus (Table 7), and the maximum duration of follow up in this study was only 3 years. We did find associations between the types of initial immunosuppressive

Table 8: Comparison of transplantation recipients included or excluded from the present analysis

Patients' characteristic	Included (n = 35 765)	Excluded (n = 40 702)	p-value (χ^2)
Age (years)			
0–17	3%	8%	<0.0001
18–34	19%	20%	0.0003
35–49	32%	35%	<0.0001
50–64	34%	33%	0.0032
65 +	12%	4%	<0.0001
Race			
White	62%	73%	<0.0001
Black	31%	20%	<0.0001
Asian	5%	5%	0.1210
Other	3%	3%	0.7143
Ethnicity			
Non-Hispanic	85%	89%	<0.0001
Hispanic	15%	11%	<0.0001
Gender			
Female	40%	41%	0.0235
Male	60%	59%	0.0235
Primary cause of kidney failure			
Diabetes	24%	19%	<0.0001
Hypertension	22%	12%	<0.0001
Glomerulonephritis	24%	24%	0.7898
Cystic kidney disease	7%	10%	<0.0001
Other	23%	35%	<0.0001
Donor type			
Living	24%	49%	<0.0001
Cadaveric	76%	51%	<0.0001
Pre–transplantation dialysis time			
Preemptive	3%	19%	<0.0001
<1 years	13%	31%	<0.0001
1 ≤ 2 years	20%	22%	<0.0001
2 ≤ 3 years	21%	10%	<0.0001
3+ years	43%	10%	<0.0001
Unknown	<1%	8%	<0.0001
Education (highest level attained)			
<College	59%	56%	<0.0001
College	10%	18%	<0.0001
Unknown	32%	27%	<0.0001
Employment			
Unable to work	39%	24%	<0.0001
Able to work	45%	60%	<0.0001
Unknown	16%	16%	0.5736
Baseline immunosuppression			
Cyclosporine	17%	15%	<0.0001
Microemulsion cyclosporine	47%	48%	0.0562
Tacrolimus	27%	28%	<0.0001
Sirolimus	6%	6%	0.4306
Azathioprine	20%	19%	<0.0001
Mycophenolate mofetil	62%	62%	0.3428
Antibody induction			
None	58%	61%	<0.0001
IL-2 receptor antibody	18%	20%	<0.0001
Other	24%	19%	<0.0001

IL–2 = interleukin-2.

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agents and the incidence of nonmelanoma skin cancer (Table 2). Specifically, tacrolimus was associated with a 35% lower incidence of skin cancer, while azathioprine was associated with a 17% higher incidence ($p = 0.04$). Antibody induction was associated with a 17% lower risk of subsequent skin cancers (Table 2). These results should be interpreted with caution, as the selection of immunosuppressive agents is often tailored to individual patients based on risk, and it is impossible to completely remove this bias in the multivariate statistical analysis. Only randomized controlled trials can determine the effects of immunosuppressive agents on outcomes.

Cancer has been reported to be more common among hemodialysis patients compared with the general population (17). Although dialysis patients do not generally receive immunosuppressive mediations, uremia is associated with abnormalities in the immune system. For this and other reasons, it is plausible that the rate of cancer after kidney transplantation may not be different than the rate of cancer in patients with end-stage kidney disease who have not undergone transplantation. Our analysis suggests that the rates of several cancers, which are increased after transplantation compared with the general population, are similar vs. comparable patients on the waiting list. However, several other tumors (e.g. skin, kidney and lymphomas) were more common after transplantation than on the waiting list (Table 5).

There is increasing evidence that viruses may play an important role in the pathogenesis of skin cancers (18,19), cervical carcinomas (20,21), Kaposi's sarcomas (22,23), hepatobiliary tumors (24,25), and lymphomas (26,27). This may help to explain why these malignancies are particularly common after kidney transplantation. The high prevalence of viral hepatitis in the dialysis population may explain why hepatobiliary tumors are higher in transplant recipients compared with the general population, but not higher compared with the waiting list. The pathogenesis of renal cell carcinoma is unclear, but it is well known that kidney cancer often occurs in end-stage kidney disease (28). The present results suggest that kidney cancer is even more common after transplantation compared with patients on the waiting list.

The incidence of nonskin cancer is higher among transplant recipients who are older, male, White (compared with Black recipients, Asians, and other ethnic groups), non-Hispanic (compared with Hispanic), and lower among patients who had diabetes as a cause of kidney disease (Table 2). These findings are in general agreement with those of another, recent study (15). It is not clear why diabetes was associated with a lower risk, although we could speculate that a lower rate of cigarette smoking among diabetics could reduce the overall risk of cancer (29). We also found that the adjusted incidence of cancer was not different in patients transplanted in 1999–2001 compared with 1995–1998 (Table 2). In another recent report the incidence of nonskin cancers increased over time (15). This analysis used data reported to the United Network for Organ Sharing, and did not adjust for the type of immunosuppressive agents used (15). It is possible that these differences, as well as differences in the populations analyzed (the current analysis included only Medicare beneficiaries), explain these different results.

A major limitation of this study (and other registry studies) is the relatively short duration of follow up (3 years). An inherent limitation of registry data comes from the fact that centers may inconsistently report events that occur late after transplantation, when patients have returned home and the transplant center can no longer follow patients closely. In the current study, we did not rely on reporting from centers to detect cancers, but rather relied on physicians, hospitals and clinics to bill for services related to the diagnosis and treatment of cancer. Nevertheless, reliance on Medicare claims limited our analysis to the first 3 years after transplantation.

Another limitation of this analysis is that the population of patients who have Medicare as their primary provider is not a random sample of the whole kidney transplant population. Thus, the results of this analysis may not be entirely applicable to the whole transplant population. Nevertheless, the Medicare primary provider population is a sizeable portion of the whole transplant population (47% in this study), and the populations analyzed in this study were similar compared with those excluded from the analysis (Table 8).

In summary, this analysis suggests that most tumors occur more frequently after kidney transplantation than in the general population, and some cancers occur more frequently than in comparable patients on the transplantation waiting list. Altogether, these results suggest cancer should continue to be a major focus of prevention in kidney transplantation.

Acknowledgment

The data reported here have been supplied by the United States Renal Data System. The interpretation and reporting of these data are the responsibility of the authors and in no way should be seen as an official policy or interpretation of the US Government.

References

1. Brunner FP, Landais P, Selwood NH. Malignancies after renal transplantation: the EDTA-ERA registry experience. European dialysis and transplantation association–European Renal Association. Nephrol Dial Transplant 1995; 10: 74–80.
2. Stewart T, Tsai S-CJ, Grayson H, Henderson R, Opelz G. Incidence of de-novo breast cancer in women chronically immunosuppressed after organ transplantation. Lancet 1995; 346: 796–798.
3. Sheil AGR. Cancer report. In: Disney, APS, ed. The Twentieth Report Australia and New Zealand Dialysis and Transplant Registry 1997. Adelaide, South Australia: ANZDATA Registry, 1997: 138–146.

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4. Birkeland SA, Lokkegaard H, Storm HH. Cancer risk in patients on dialysis and after renal transplantation. Lancet 2000; 355: 1886–1887.
5. Adami J, Gabel H, Lindelof B et al. Cancer risk following organ transplantation: a nationwide cohort study in Sweden. Br J Cancer 2003; 89: 1221–1227.
6. United States Renal Data System. USRDS 2003 Annual Data Report. Atas of End-Stage Renal Disease in the United States. Bethesda, MD: National Institutes of Health. National Institute of Diabetes and Digestive Kidney Diseases, 2003.
7. Hebert PL, Geiss LS, Tierney EF, Engelgau MM, Yawn BP, McBean AM. Identifying persons with diabetes using Medicare claims data. Am J Med Qual 1999; 14: 270–277.
8. United States Cancer Statistics Working Group. United States Cancer Statistics: 1999 Incidence. Atlanta, GA: Department of Health and Human Services. Centers for Disease Control and Prevention and National Cancer Insitute, 2002.
9. US Preventative Services Task Force. Screening for prostate cancer: recommendation and rationale. Ann Intern Med 2002; 137: 915–916.
10. US Preventative Services Task Force. Screening for colorectal cancer: recommendation and rationale. Ann Intern Med 2002; 137: 129–131.
11. US Preventative Services Task Force. Screening for breast cancer: recommendations and rationale. Ann Intern Med 2002; 137: 344–346.
12. The Tobacco Use Dependence Clinical Practice Guideline Panel, Staff, Consortium Representatives. A clinical practice guideline for treating tobacco use and dependence: a US Public Health Service report. JAMA 2000; 283: 3244–3254.
13. Kiberd BA, Keough-Ryan T, Clase CM. Screening for prostate, breast and colorectal cancer in renal transplant recipients. Am J Transplant 2003; 3: 619–625.
14. Opelz G, Döhler B. Lymphomas after solid organ transplantation: a collaborative transplant study report. Am J Transplant 2003; 4: 222–230.
15. Bustami RT, Ojo AO, Wolfe RA et al. Immunosuppression and the risk of post-transplant malignancy among cadaveric first kidney transplant recipients. Am J Transplant 2004; 4: 87–93.
16. Luan FL, Ding R, Sharma VK, Chon WJ, Lagman M, Suthanthiran M. Rapamycin is an effective inhibitor of human renal cancer metastasis. Kidney Int 2003; 63: 917–926.
17. Maisonneuve P, Agodoa L, Gellert R et al. Cancer in patients on dialysis for end-stage renal disease: an international collaborative study. Lancet 1999; 354: 93–99.
18. Jong-Tieben LM, Berkhout RJ, ter Schegget J et al. The prevalence of human papillomavirus DNA in benign keratotic skin lesions of renal transplant recipients with and without a history of skin cancer is equally high: a clinical study to assess risk factors for keratotic skin lesions and skin cancer. Transplantation 2000; 69: 44–49.
19. Harwood CA, Surentheran T, McGregor JM et al. Human papillomavirus infection and non-melanoma skin cancer in immunosuppressed and immunocompetent individuals. J Med Virol 2000; 61: 289–297.
20. Nobbenhuis MA, Walboomers JM, Helmerhorst TJ et al. Relation of human papillomavirus status to cervical lesions and consequences for cervical-cancer screening: a prospective study. Lancet 1999; 354: 20–25.
21. Clifford GM, Smith JS, Plummer M, Munoz N, Franceschi S. Human papillomavirus types in invasive cervical cancer worldwide: a meta-analysis. Br J Cancer 2003; 88: 63–73.
22. Moosa MR, Treurnicht FK, van Rensburg EJ, Schneider JW, Jordaan HF, Engelbrecht S. Detection and subtyping of human herpesvirus-8 in renal transplant patients before and after remission of Kaposi's sarcoma. Transplantation 1998; 66: 214–218.
23. Farge D, Lebbe C, Marjanovic Z et al. Human herpes virus-8 and other risk factors for Kaposi's sarcoma in kidney transplant recipients. Groupe Cooperatif de Transplantation d'Ile de France (GCIF). Transplantation 1999; 67: 1236–1242.
24. Monto A, Wright TL. The epidemiology and prevention of hepatocellular carcinoma. Semin Oncol 2001; 28: 441–449.
25. El Serag HB. Hepatocellular carcinoma: an epidemiologic view. J Clin Gastroenterol 2002; 35: S72–S78.
26. Paya CV, Fung JJ, Nalesnik MA et al. Epstein-Barr virus-induced posttransplant lymphoproliferative disorders. ASTS/ASTP EBV-PTLD Task Force and The Mayo Clinic Organized International Consensus Development Meeting. Transplantation 1999; 68: 1517–1525.
27. Meijer E, Dekker AW, Weersink AJ, Rozenberg-Arska M, Verdonck LF. Prevention and treatment of Epstein-Barr virus-associated lymphoproliferative disorders in recipients of bone marrow and solid organ transplants. Br J Haematol 2002; 119: 596–607.
28. Stewart JH, Buccianti G, Agodoa L et al. Cancers of the kidney and urinary tract in patients on dialysis for end-stage renal disease: analysis of data from the United States, Europe, and Australia and New Zealand. J Am Soc Nephrol 2003; 14: 197–207.
29. Kasiske BL, Klinger D. Cigarette smoking in renal transplant recipients. J Am Soc Nephrol 2000; 11: 753–759.

Appendix

ICD-9-CM codes used to identify specific cancers

Skin	140.x, 173.x, 187.x
Mouth	141.x, 142.x, 143.x, 144.x, 145.x, 146.x, 147.x, 148.x, 149.x
Esophagus	150.x
Stomach	151.x
Small intestine	152.x
Colon	153.x, 154.x
Hepatobiliary	155.x, 156.x
Pancreas	157.x
Larynx	161.x
Lung	162.x, 163.x
Bone	170.x
Breast (female)	174.x
Breast (male)	175.x
Kaposi's sarcoma	176.x
Melanoma	172.x
Uterus	179.x, 182.x
Cervix	180.x
Ovary	183.x
Vulvovaginal	184.x
Prostate	185.x
Testes	186.x
Bladder	188.x
Kidney	189.x
CNS	191.x, 192.x
Endocrine	193.x, 194.x
Lymphoma	200.x, 202.x, 204.x
Hodgkin's	201.x
Myeloma	203.x
Leukemia	205.x, 206.x, 207.x, 208.x
Other	158.x, 159.x, 160.x, 164.x, 165.x 171.x, 181.x, 190.x, 195.x, 196.x, 197.x, 198.x, 199.x

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